Project Design Phase-I
Solution Architecture

Date	19 September 2022
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - xxx
Maximum Marks	4 Marks

Solution Architecture:

Solution Architecture Overview

Solution architecture explains how technology can be used to solve a specific problem. In this project, the architecture focuses on analyzing heart disease data and presenting useful insights through interactive dashboards. The data is collected, stored in a database, processed using SQL queries, and visualized using Tableau. These dashboards are then integrated into a web interface using Flask so that users can easily access and interact with the visualizations.

Architecture Layers:

1. Data Collection Layer

The first step in the system is collecting the heart disease dataset from reliable healthcare data sources. The dataset includes information such as age category, gender, BMI, smoking status, alcohol consumption, physical activity, and general health condition. This data is important for analyzing lifestyle factors related to heart disease.

2. Database Layer

After collecting the data, it is stored in a relational database like **MySQL or PostgreSQL**. The database organizes the data in a structured way and allows efficient storage and retrieval of information when needed for analysis.

3. Data Processing Layer

In this stage, SQL queries are used to clean and prepare the dataset. Operations such as filtering unnecessary data, correcting data formats, and organizing the dataset are performed so that it becomes suitable for analysis and visualization.

4. Visualization Layer

Tableau is used to create visualizations such as bar charts, scatter plots, and heat maps. These visual tools help users understand patterns and relationships between different health factors and heart disease risk.

5. Web Integration Layer

The Tableau dashboards are integrated into a Flask-based web application. This allows users to access the dashboards through a web interface and interact with the visualizations easily.

Data Flow

The working process of the system can be described in the following steps:

1. Heart disease data is collected from healthcare sources.
2. The dataset is stored in a structured SQL database.
3. SQL queries are used to clean and prepare the data.
4. Tableau connects to the database and creates visualizations.
5. Interactive dashboards are developed in Tableau.
6. These dashboards are embedded into a Flask web interface where users can explore the insights.

Example - Solution Architecture Diagram:

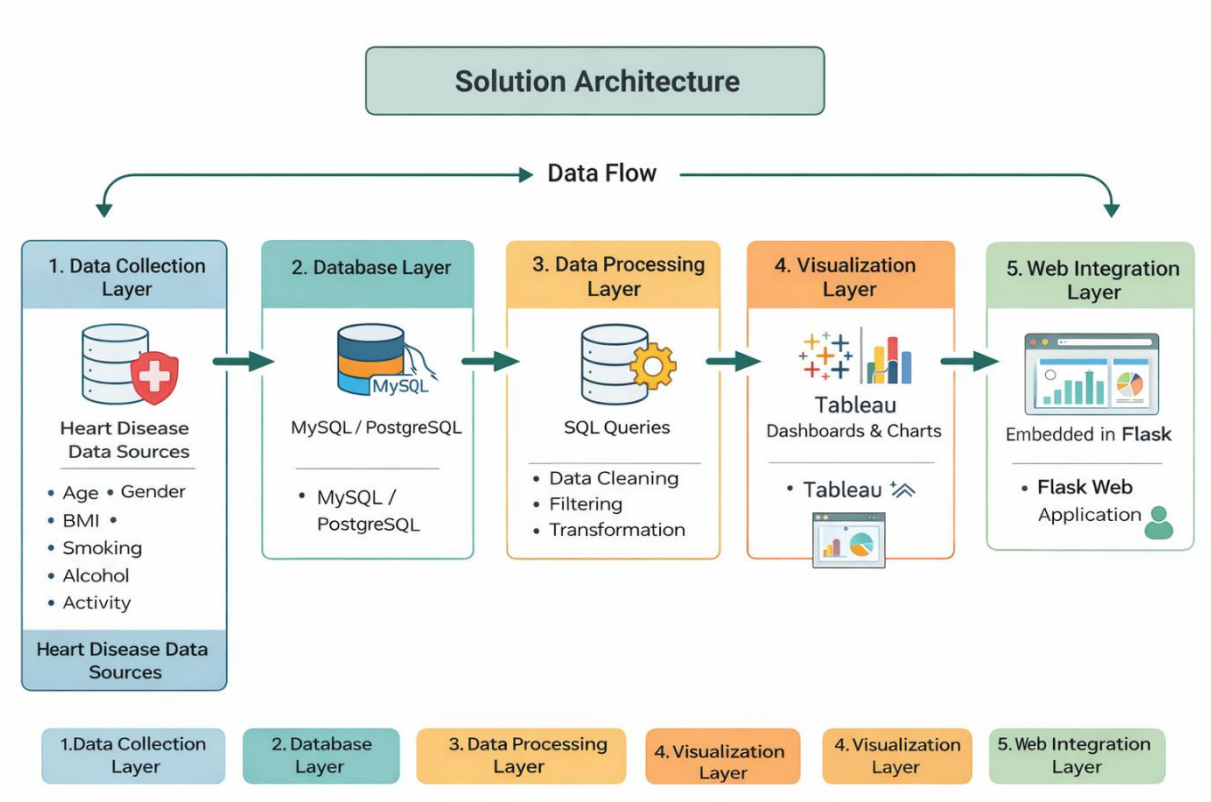

Figure 1: Architecture and data flow of the Heart Disease Analysis

Reference: https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view